

PART-A

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Signature of the Invigilator: [Signature]

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	3	9	17

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

we know for LSE we need to minimize

$$Z = \sum_{i=1}^n e_i^2 = \sum_{i=1}^3 e_i^2$$

$$= (y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2$$

$$\frac{\partial Z}{\partial \theta_1} = -2(y_1 - \theta_1 - \theta_2) - 2(y_2 - \theta_1 + 2\theta_2) - 4(y_3 - 2\theta_1 + \theta_2)$$

$$\frac{\partial Z}{\partial \theta_2} = -2(y_1 - \theta_1 - \theta_2) + 4(y_2 - \theta_1 + 2\theta_2) + 2(y_3 - 2\theta_1 + \theta_2)$$

As both $\frac{\partial Z}{\partial \theta_1} > 0$ & $\frac{\partial Z}{\partial \theta_2} > 0$

So min will occur when

$$\frac{\partial Z}{\partial \theta_1} = 0 \quad \& \quad \frac{\partial Z}{\partial \theta_2} = 0.$$

$$\Rightarrow (y_1 - \hat{\theta}_1 - \hat{\theta}_2) + (y_2 - \hat{\theta}_1 + 2\hat{\theta}_2) + 2(y_3 - 2\hat{\theta}_1 + \hat{\theta}_2) = 0$$

$$\& \quad -(y_1 - \hat{\theta}_1 - \hat{\theta}_2) + 2(y_2 - \hat{\theta}_1 + 2\hat{\theta}_2) + (y_3 - 2\hat{\theta}_1 + \hat{\theta}_2) = 0.$$

$$y_1 + y_2 + 2y_3 - 6\hat{\theta}_1 + 3\hat{\theta}_2 = 0$$

$$-y_1 + 2y_2 + y_3 - 3\hat{\theta}_1 + 6\hat{\theta}_2 = 0$$

$$\Rightarrow \hat{\theta}_1 = \frac{+3y_1 + 3y_3}{9} = \frac{y_1 + y_3}{3}$$

$$\Rightarrow \hat{\theta}_2 = \frac{y_1 - y_2}{3}$$

So LE of $\hat{\theta}_1 = \frac{y_1 + y_3}{3}$

$$\hat{\theta}_2 = \frac{y_1 - y_2}{3}$$

(05)

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

(a) ~~We know Covariance matrix is~~

$$= \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \text{Cov}(X_1, X_3) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \text{Cov}(X_2, X_3) \\ \text{Cov}(X_3, X_1) & \text{Cov}(X_3, X_2) & \text{Cov}(X_3, X_3) \end{bmatrix}$$

We know Correlation matrix of $\mathbf{X}$ will be

$$\rho = \begin{bmatrix} \text{Corr}(X_1, X_1) & \text{Corr}(X_1, X_2) & \text{Corr}(X_1, X_3) \\ \text{Corr}(X_2, X_1) & \text{Corr}(X_2, X_2) & \text{Corr}(X_2, X_3) \\ \text{Corr}(X_3, X_1) & \text{Corr}(X_3, X_2) & \text{Corr}(X_3, X_3) \end{bmatrix}$$

$$\text{Now, } \text{Corr}(X_i, X_i) = \frac{\text{Cov}(X_i, X_i)}{\sqrt{\text{Var}(X_i)} \sqrt{\text{Var}(X_i)}} = \frac{\text{Var}(X_i)}{\text{Var}(X_i)} = 1$$

$$\text{e.g. } \text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)} \sqrt{\text{Var}(X_2)}}$$

$$\text{Var}(X_1) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}, \quad \text{Var}(X_2) = \text{Var}(U_2) = \frac{1}{12}$$

$$\text{Var}(X_3) = \text{Var}(U_1 + U_2) = \text{Var}(U_1) + \text{Var}(U_2) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$\text{Cov}(X_1, X_2) = \text{Cov}(U_1, U_2) = 0$$

$$\text{Cov}(X_1, X_3) = \text{Cov}(U_1, U_1 + U_2) = \text{Cov}(U_1, U_1) = \frac{1}{12}$$

$$\text{Cov}(X_2, X_3) = \text{Cov}(U_2, U_1 + U_2) = \text{Cov}(U_2, U_2) = \frac{1}{12}$$

$$\Rightarrow \text{corr}(x_1, x_2) = 0, \quad \text{corr}(x_1, x_3) = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \times \sqrt{\frac{1}{6}}} = \frac{1}{\sqrt{2}}$$

$$\text{corr}(x_2, x_3) = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \times \sqrt{\frac{1}{6}}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow P = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

(b) Variance of First principal component based on above matrix will be largest eigen value of P.

$$\Rightarrow \text{Now, } (P - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^3 - 2(1-\lambda) = 0$$

$$\Rightarrow (1-\lambda)((1-\lambda)^2 - 2) = 0$$

$$\Rightarrow \lambda = 1, 0, 2$$

So variance of 1st principal component is 2.

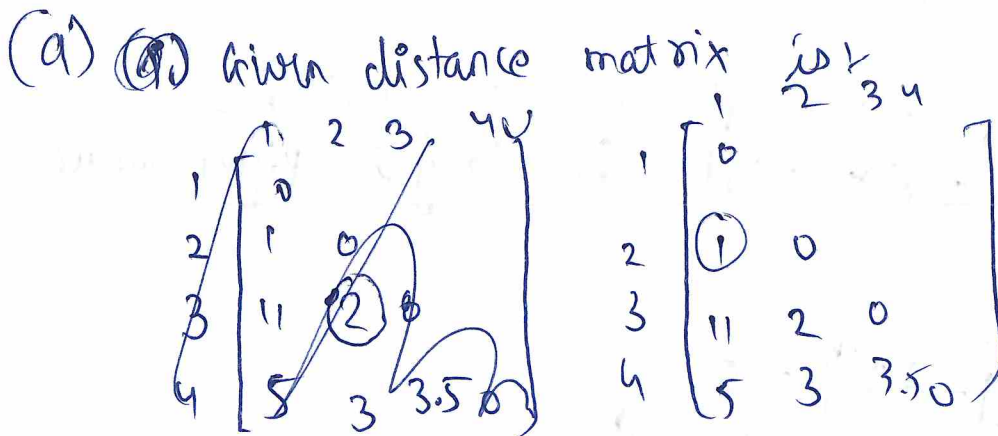
3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

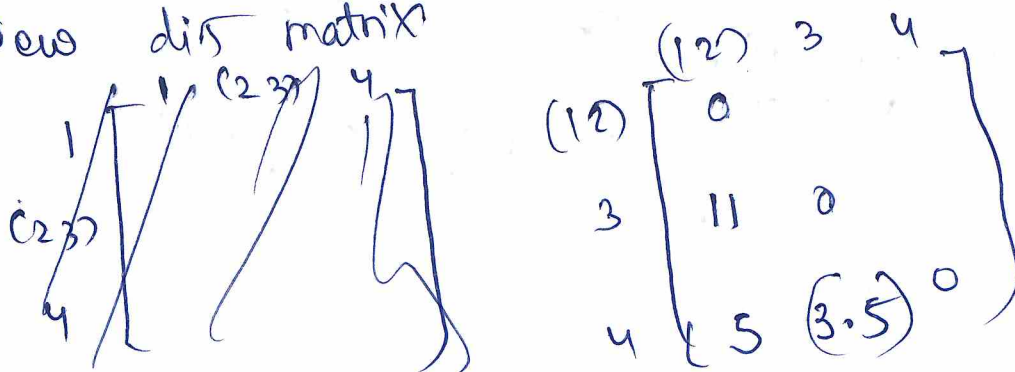
(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.



(i) Now here min distance is between 1 & 2. So
 $\rightarrow$ ~~cluster them~~ combine them. at distance = 1.

For complete linkage we take new distance
 $d_{(12)w} = \max\{d_{1w}, d_{2w}\}$

$\Rightarrow$ New dis matrix



Now min dis is between 3 & 4. So
 combine 3 & 4. at distance = 3.5

$\Rightarrow$ New dist matrix

$$\begin{matrix} (12) \\ (34) \end{matrix} \begin{bmatrix} (12) & (34) \\ 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} (12) & (34) \\ (12) & (34) \end{matrix} \begin{bmatrix} 0 & 0 \end{bmatrix}$$

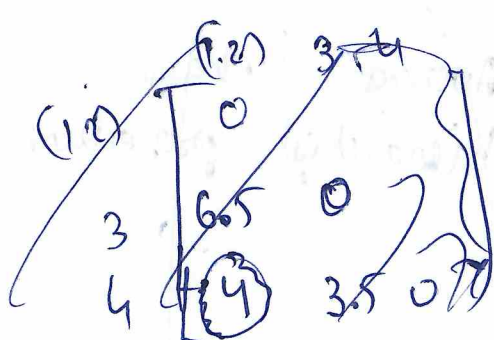
Now only remained is combining (12) & (34)
 $\Rightarrow ((12) (34))$ at distance 211.

(ii) For average linkage.

$$d(u,v,w) = \frac{\sum_{i=1}^N \sum_{j=1}^N d_{ij}}{N_u \times N_v}$$

Now first we combine 2, 1 at distance 1.

$d_{12} = 0, d_{13} = 6, d_{14} = 4, d_{23} = 3, d_{24} = 3.5, d_{34} = 0$
 $d_{123} = \frac{d_{13} + d_{23}}{2} = 6.5$
 $d_{124} = \frac{d_{14} + d_{24}}{2} = 4$



$$\begin{matrix} (12) & 3 & 4 \\ (12) & \begin{bmatrix} 0 & 3.5 & 0 \\ 6.5 & 0 & 0 \\ 4 & 3.5 & 0 \end{bmatrix} \end{matrix}$$

min here is 4 so combine (1,2) with 4.
 $d_{1243} = \frac{d_{13} + d_{23} + d_{43}}{3} = \frac{6 + 3 + 3.5}{3} = 5.5$

ADDITIONAL PAGES FOR ANSWERS

~~So new dist matrix is~~

$$\begin{matrix} & (12) & 3 \\ (12) & \begin{bmatrix} 0 & 3 \\ 3 & 5.5 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} & (1234) \\ (1234) & \begin{bmatrix} 0 & 5.5 \end{bmatrix} \end{matrix}$$

~~combine (1, 2, 3, 4) forming full cluster.~~

now min distn is 3-5 between 3, 4.
combine them

$$d_{12}(34) = \frac{d_{13} + d_{23} + d_{14} + d_{24}}{4 \times 2}$$

$$= \frac{21}{4} = 5.25$$

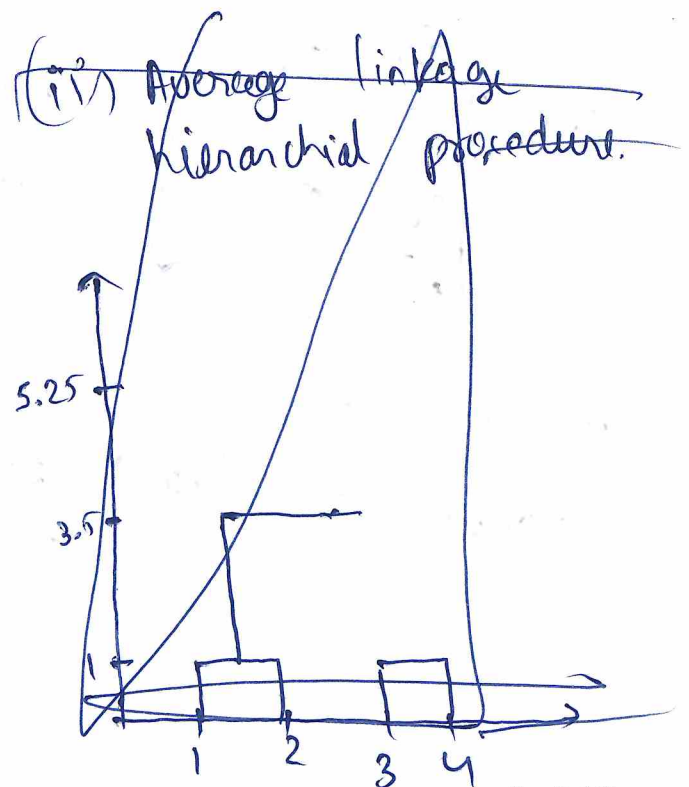
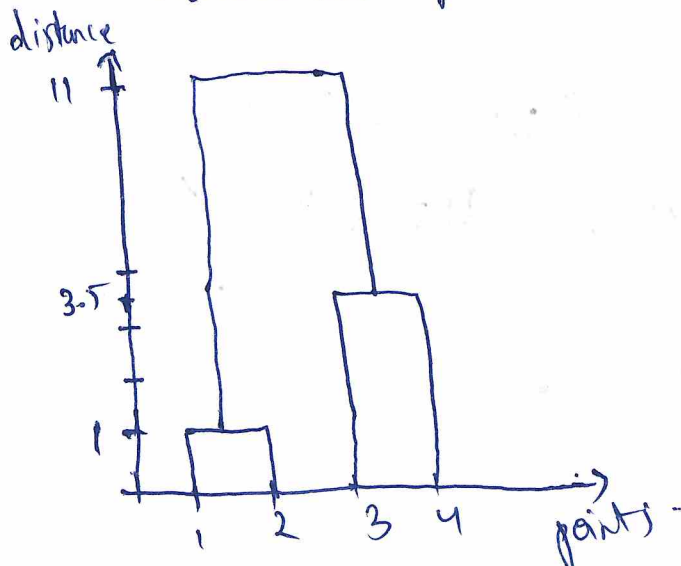
Ob

$$\begin{matrix} (12) & (34) \\ (12) & \begin{bmatrix} 0 & 5.25 \\ 5.25 & 0 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} & ((12)(34)) \\ ((12)(34)) & \begin{bmatrix} 0 \end{bmatrix} \end{matrix}$$

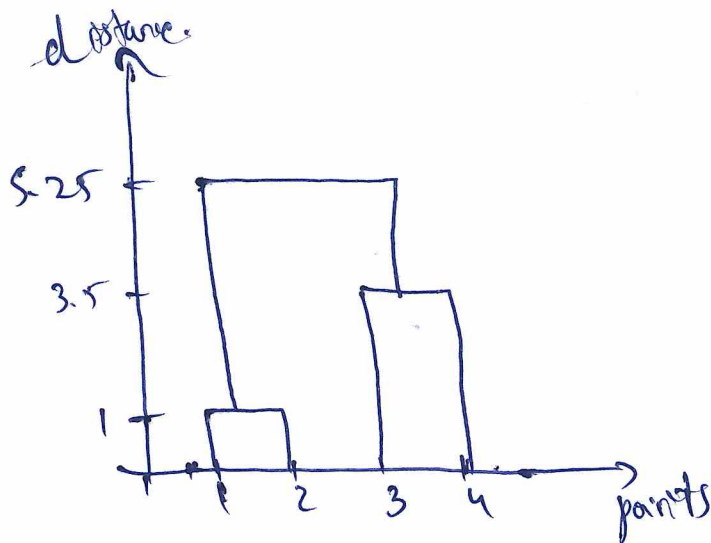
only remained combination. is (12) with (34).

(b) Dendrograms.

(i) Complete linkage hierarchical procedure



(17) Average linkage hierarchical procedure



We can see that complete linkage displays combines ~~at distance~~, Average linkage ~~because~~ ~~max~~ ~~average~~

In complete linkage.

First ~~(1,2)~~ 1 combine with 2 at 1
3 with 4 at 3.5
an then (12) with (34) at 11.

Whereas in average linkage it combines

First ~~(1,2)~~ 1 with 2 at 1

3 with 4 at 3.5

(12) with (34) at 5.25 $\neq$ complete linkage combine dist. once

03

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by



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$f(x) = \begin{cases} x^2 & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$

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4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by